EI331 Signals and Systems

Lecture 11

Bo Jiang

John Hopcroft Center for Computer Science
Shanghai Jiao Tong University

April 2, 2019

Contents

1. Mean-square Convergence of Fourier Series

2. Pointwise Convergence of Fourier Series

3. Fourier Series of Periodic Impulse Train

4. Filtering

Best Mean-square Approximation

Given orthonormal system $\{e_1, \ldots, e_n\}$, best mean-square approximation of x by element in span$\{e_1, \ldots, e_n\}$, i.e. by element of form $y = \sum_{k=1}^{n} a_k e_k$ is

$$y = \sum_{k=1}^{n} \langle x, e_k \rangle e_k,$$

with minimum mean-square error

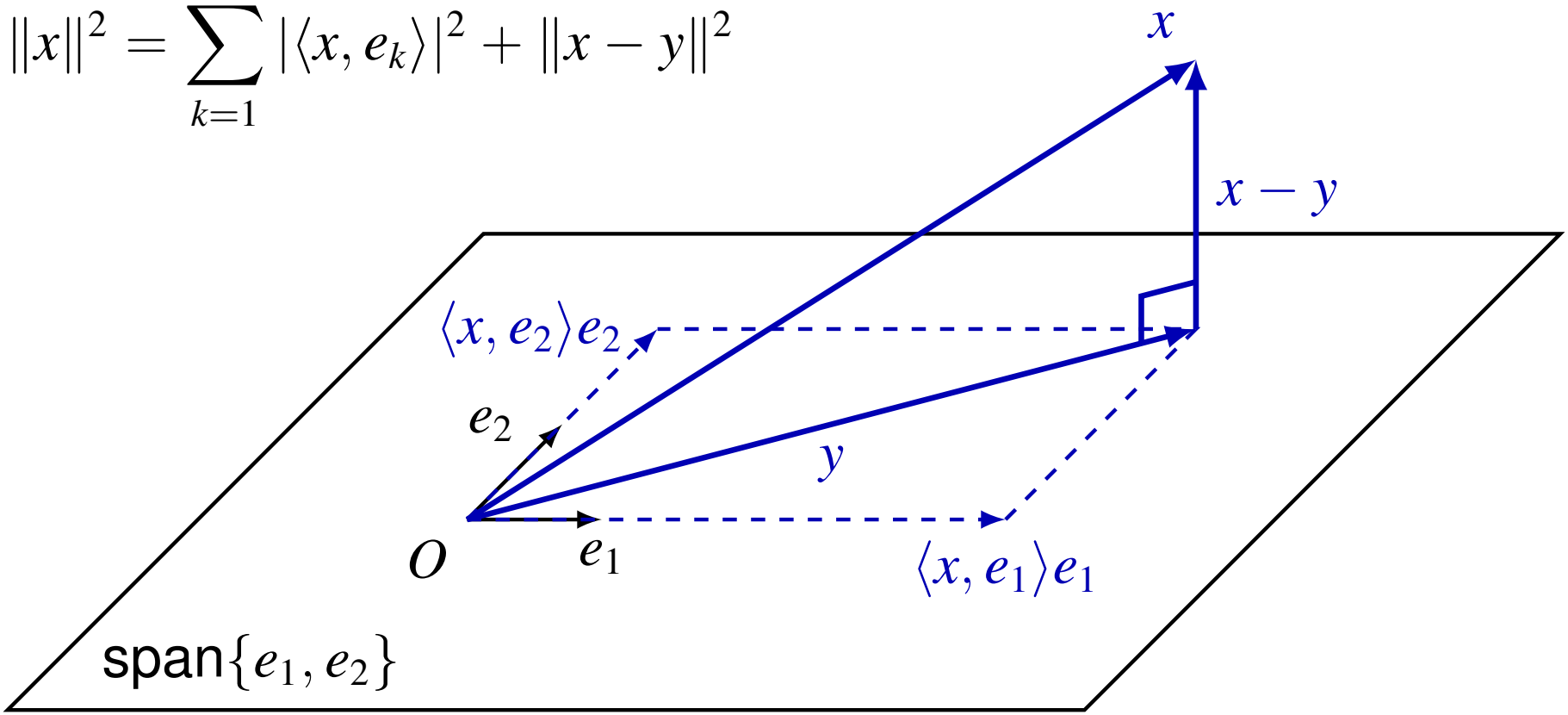
$$\|x - y\|^2 = \|x\|^2 - \sum_{k=1}^{n} |\langle x, e_k \rangle|^2.$$

Geometry of Best Mean-square Approximation

Orthogonal projection onto subspace span$\{e_1, \ldots, e_n\}$

Pythagorean theorem

$$\|x\|^2 = \sum_{k=1}^{n} |\langle x, e_k \rangle|^2 + \|x - y\|^2$$



Convergence in Normed Vector Space

In normed vector space $(V, \|\cdot\|)$, sequence $\{x_n\}$ converges to $x \in V$ if

$$\lim_{n\to\infty} \|x_n - x\| = 0$$

We say x is the limit of $\{x_n\}$ and write

$$x = \lim_{n\to\infty} x_n, \quad \text{or} \quad x_n \to x, \text{ as } n \to \infty.$$

Sequence $\{x_n\} \subset V$ is Cauchy sequence iff

$$\|x_n - x_m\| \to 0, \qquad \text{as } n, m \to \infty.$$

Theorem. Every convergent sequence is Cauchy.

Proof. If $x_n \to x$, triangle inequality yields

$$\|x_n - x_m\| \le \|x_n - x\| + \|x_m - x\| \to 0, \qquad \text{as } n, m \to \infty.$$

Banach Space

Normed vector space $(V, \|\cdot\|)$ is complete if every Cauchy sequence converges.

Complete normed vector space is called Banach space.

Examples of Banach spaces

- $(\mathbb{R}^n, \|\cdot\|_p)$, $(\mathbb{C}^n, \|\cdot\|_p)$
- DT signal space $\ell_p = \{x \in \mathbb{C}^{\mathbb{Z}} : \|x\|_p < \infty\}$ with ℓ_p norm
- CT signal space $L_p = \{x \in \mathbb{C}^{\mathbb{R}} : \|x\|_p < \infty\}$ with L_p norm
- Space $L_2(T)$ of CT signals with period T and finite average power, $\|x\|^2 = \frac{1}{T}\int_T |x(t)|^2 dt < \infty$
- Space $C(T)$ of continuous T-periodic CT signals with L_∞ norm

$$\|x\|_\infty = \sup_{t\in[0,T]} |x(t)| = \sup_{t\in\mathbb{R}} |x(t)|.$$

$x_n \to x$ in L_∞ norm means uniform convergence

Banach Space

Examples of incomplete spaces

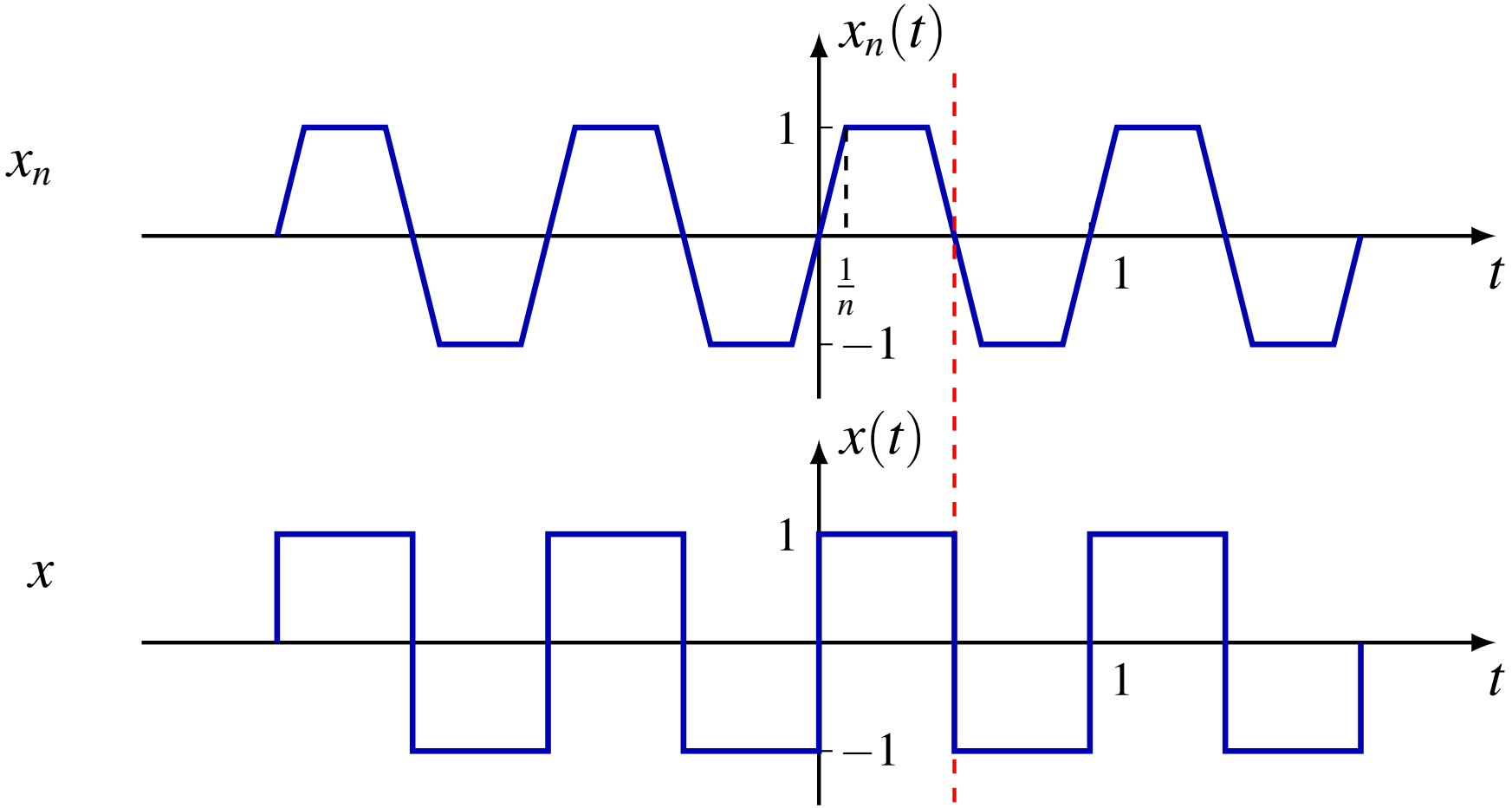
- $(\mathbb{Q}, |\cdot|)$. Its completion is $(\mathbb{R}, |\cdot|)$
- Space $C(T)$ of continuous T-periodic CT signals with L_2 norm. Its completion is $L_2(T)$
  - periodic odd function with period $T = 1$

$$x_n = \begin{cases} nt, & 0 \le t \le \frac{1}{n} \\ 1, & \frac{1}{n} \le \frac{1}{2} - \frac{1}{n} \\ 1 - nt, & \frac{1}{2} - \frac{1}{n} \le t \le \frac{1}{2} \end{cases}$$

  - $x_n \in C(1)$
  - x_n converges to periodic square wave in L_2 norm
  - but periodic square function **not** in $C(1)$

Same vector space with different norms becomes **different** normed vector spaces, e.g $(C(T), \|\cdot\|_\infty)$ vs. $(C(T), \|\cdot\|_2)$

Incompleteness of $C(T)$ with L_2 Norm



$$\|x_n - x\|_2^2 = \left(\int_{-\frac{1}{n}}^{\frac{1}{n}} + \int_{\frac{1}{2}-\frac{1}{n}}^{\frac{1}{2}+\frac{1}{n}} \right) |x_n(t) - x(t)|^2 dt \leq \frac{4}{n} \to 0$$

$x \in L_2(1)$ but $x \notin C(1)$. Also $\|x_n - x\|_\infty = 1, \forall n$

L_2 Convergence vs. Pointwise Convergence

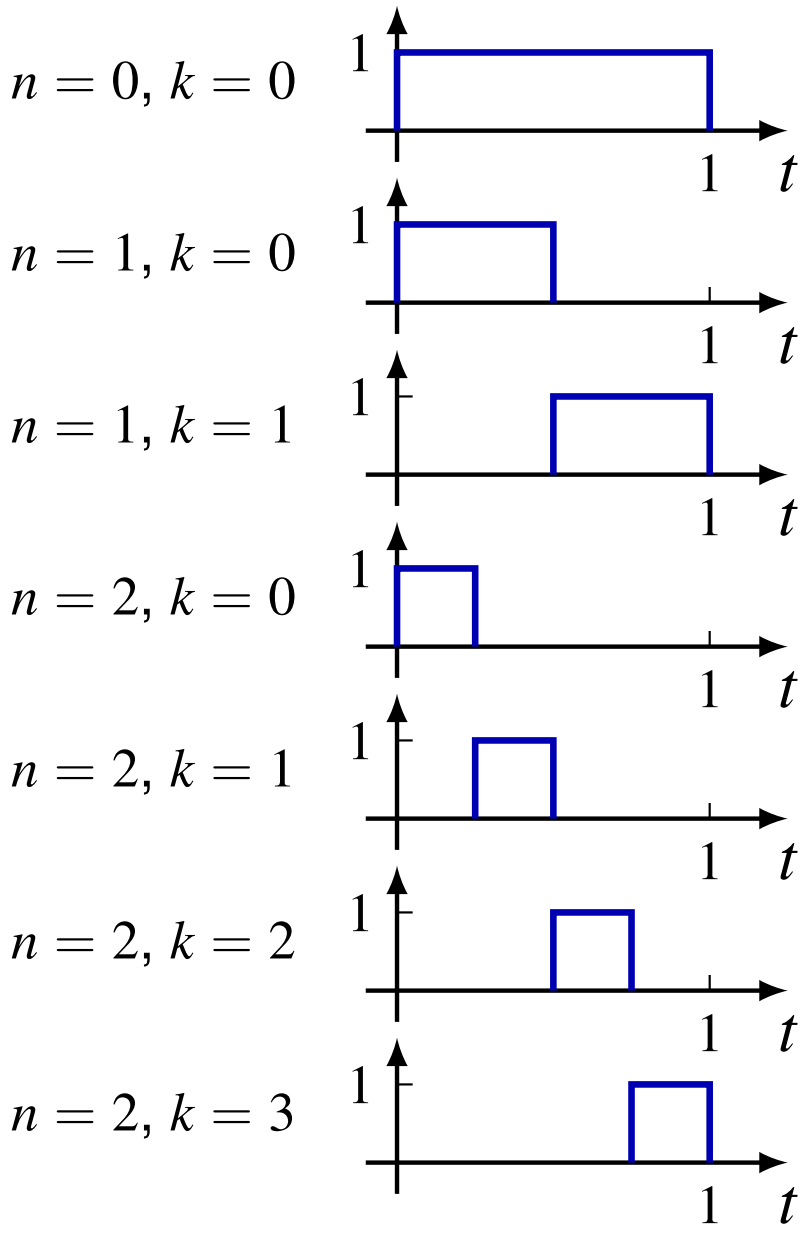
For $n \geq 0$ and $0 \leq k < 2^n$,

$$x_{2^n+k} = \begin{cases} 1, & \frac{k}{2^n} \leq t < \frac{k+1}{2^n} \\ 0, & \text{otherwise} \end{cases}$$

- $x_{2^n+k} \to 0$ in L_2 norm

$$\|x_{2^n+k} - 0\|_2 = \frac{1}{2^{n/2}} \to 0$$

- **not** convergent at any $t_0 \in [0, 1)$
  - $\forall n, \exists k_1$ s.t $x_{2^n+k_1}(t_0) = 1$
  - $\forall n, \exists k_0$ s.t $x_{2^n+k_0}(t_0) = 0$



Hilbert Space

Inner product space $(V, \langle \cdot, \cdot \rangle)$ is also normed vector space with induced norm $\|x\| = \sqrt{\langle x, x \rangle}$.

$(V, \langle \cdot, \cdot \rangle)$ is complete if $(V, \| \cdot \|)$ with induced norm is complete.

Complete inner produce space is called Hilbert space.

Examples of Hilbert spaces

- $\mathbb{R}^n$ with $\langle x, y \rangle = \sum_{k=1}^{n} x_k y_k$; $\mathbb{C}^n$ with $\langle x, y \rangle = \sum_{k=1}^{n} x_k \bar{y}_k$
- Space ℓ_2 with $\langle x, y \rangle = \sum_{k=-\infty}^{\infty} x_k \bar{y}_k$
- Space L_2 with $\langle x, y \rangle = \int_{\mathbb{R}} x(t)\overline{y(t)}dt$
- Space $L_2(T)$ with $\langle x, y \rangle = \frac{1}{T} \int_T x(t)\overline{y(t)}dt$

Example of incomplete space

- Space $C(T)$ with $\langle x, y \rangle = \frac{1}{T} \int_T x(t)\overline{y(t)}dt$
  $L_2(T)$ is completion of $C(T)$.

Complete Orthonormal Sequence

Orthonormal sequence $\{e_k : k \in \mathbb{N}\}$ in Hilbert space H is complete if **every** $x \in H$ has expansion

$$x = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k$$

i.e.

$$\lim_{n \to \infty} \|x - \sum_{k=1}^{n} \langle x, e_k \rangle e_k\| = 0$$

Complete orthonormal sequence also called orthonormal basis of H.

Example. $\{\delta_k : k \in \mathbb{Z}\}$ is orthonormal basis of ℓ_2

Parseval's Identity

If $\{e_k : k \in \mathbb{N}\}$ orthonormal basis of H,

$$\|x\|^2 = \sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2$$

Proof. By Pythagorean theorem

$$\|x\|^2 = \sum_{k=1}^{n} |\langle x, e_k \rangle|^2 + \|x - \sum_{k=1}^{n} \langle x, e_k \rangle e_k\|^2$$

Let $n \to \infty$ and last term goes to zero by definition of orthonormal basis.

Parseval's Identity

H is isomorphic to ℓ_2, $x \leftrightarrow \{\langle x, e_k\rangle : k \in \mathbb{N}\}$ and

$$\langle x, y\rangle = \sum_{k=1}^{\infty} \langle x, e_k\rangle \overline{\langle y, e_k\rangle}$$

Proof.

$$\begin{aligned}
\langle x, y\rangle &= \langle \sum_k \langle x, e_k\rangle e_k, \sum_m \langle y, e_m\rangle e_m\rangle \\
&= \sum_k \langle x, e_k\rangle \sum_m \overline{\langle y, e_m\rangle} \langle e_k, e_m\rangle \\
&= \sum_k \langle x, e_k\rangle \sum_m \overline{\langle y, e_m\rangle} \delta[k-m] \\
&= \sum_k \langle x, e_k\rangle \overline{\langle y, e_k\rangle}
\end{aligned}$$

NB. When $x = y$, we recover $\|x\|^2 = \sum_{k=1}^{\infty} |\langle x, e_k\rangle|^2$

Mean-square Convergence of Fourier Series

Theorem. $\{e^{jk\frac{2\pi}{T}t} : k \in \mathbb{Z}\}$ is orthonormal basis of $L_2(T)$.
For any $x \in L_2(T)$, Fourier series converges in mean-square, i.e. in L_2 norm

$$\lim_{N\to\infty} \|x - S_N(x)\| = 0$$

NB. Convergence in mean-square does **not** imply pointwise convergence

Parseval's identity

$$\sum_{k=-\infty}^{\infty} |\hat{x}[k]|^2 = \|x\|^2 = \frac{1}{T}\int_T |x(t)|^2 dt$$

Interpretation: Energy conservation

- $|\hat{x}[k]|^2$ is average power of k-th harmonic component
- $\|x\|^2$ is average power of x

Fourier Series of L_2 Signals

Correspondence between periodic functions and doubly infinite sequences; time domain vs. frequency domain

$$x \xleftrightarrow{\mathcal{FS}} \hat{x} \qquad \text{or} \qquad x(t) \xleftrightarrow{\mathcal{FS}} \hat{x}[k]$$

Fourier series is isomorphism between Hilbert space of L_2 signals and Hilbert space of ℓ_2 Fourier coefficients

$$\begin{aligned} \mathcal{FS} : L_2(T) &\to \ell_2 \\ x &\mapsto \hat{x} \end{aligned}$$

Will see discrete-time Fourier transform goes from $\hat{x}$ to x

Parseval's theorem

$$\langle x, y \rangle_{L_2(T)} = \langle \hat{x}, \hat{y} \rangle_{\ell_2}$$

Contents

1. Mean-square Convergence of Fourier Series

2. Pointwise Convergence of Fourier Series

3. Fourier Series of Periodic Impulse Train

4. Filtering

Integral Representation of Partial Sum

WLOG, focus on $T = 2\pi$ and $\omega_0 = 1$

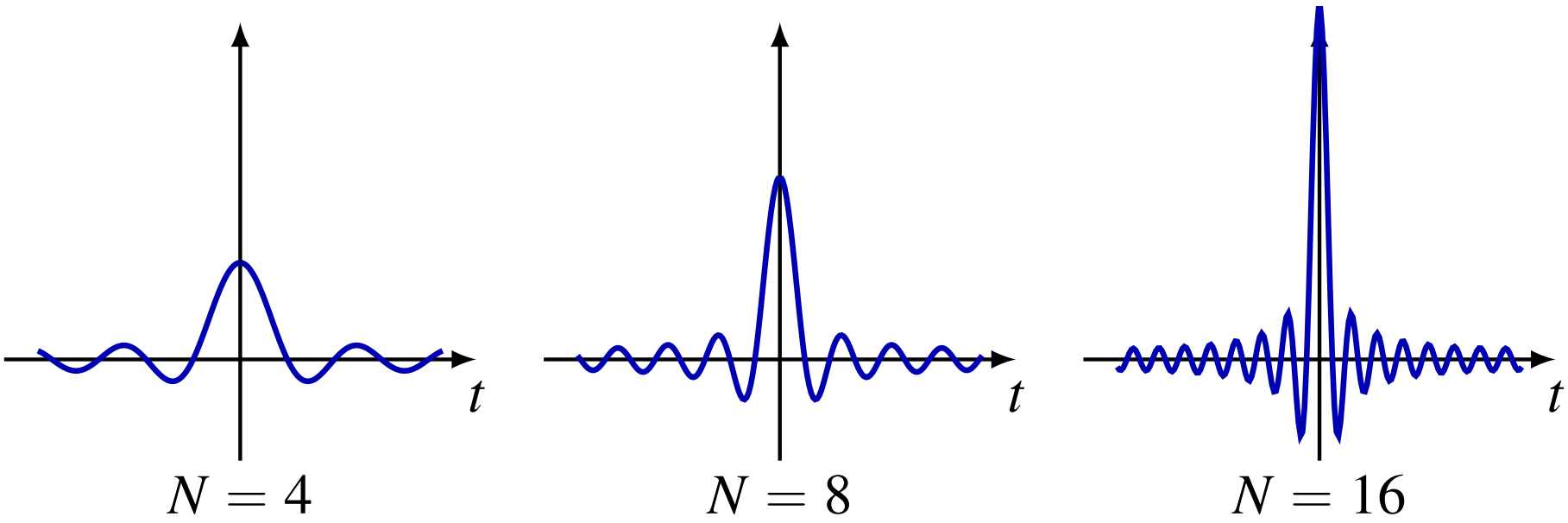
$$
\begin{aligned}
S_N(x)(t) &= \sum_{k=-N}^{N} \hat{x}[k] e^{jkt} \\
&= \sum_{k=-N}^{N} \left(\frac{1}{2\pi} \int_{2\pi} x(\tau) e^{-jk\tau} d\tau \right) e^{jkt} \\
&= \frac{1}{2\pi} \int_{2\pi} x(\tau) \sum_{k=-N}^{N} e^{jk(t-\tau)} d\tau \\
&= \frac{1}{2\pi} \int_{2\pi} x(\tau) D_N(t-\tau) d\tau = (x * \frac{D_N}{2\pi})(t)
\end{aligned}
$$

where D_N is Dirichlet kernel

$$
D_N(t) = \sum_{k=-N}^{N} e^{jkt} = \frac{\sin((N+\frac{1}{2})t)}{\sin(t/2)}
$$

Dirichlet Kernel

Plot of $\frac{D_N}{2\pi}$ for various N



If $\frac{D_N}{2\pi} \to \delta$ (more precisely, periodic impulse train on slide 22), then would have $S_N(x) = x * \frac{D_N}{2\pi} \to x * \delta = x$ at least for continuous x

Unfortunately, exists continuous x whose Fourier partial sum $S_N(x)$, as $N \to \infty$, fails to converge at all at some t, let alone converges to $x(t)$ at such t

Pointwise Convergence

Theorem. If x is differentiable, then $\lim_{N\to\infty} S_N(x)(t) = x(t)$, $\forall t$.

Proof. Fix t. Define

$$F(\tau) = \begin{cases} \dfrac{x(t-\tau) - x(t)}{\sin(\tau/2)}, & \tau \neq 0 \\ -2x'(t), & \tau = 0 \end{cases}$$

which is continuous on $[-\pi, \pi]$. Note $\frac{1}{2\pi}\int_{-\pi}^{\pi} D_N(\tau)d\tau = 1$.

$$\begin{aligned} S_N(x)(t) - x(t) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} x(t-\tau)D_N(\tau)d\tau - \frac{1}{2\pi}\int_{-\pi}^{\pi} x(t)D_N(\tau)d\tau \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} [x(t-\tau) - x(t)]D_N(\tau)d\tau \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} F(\tau)\sin\left((N+\frac{1}{2})\tau\right)d\tau \end{aligned}$$

Pointwise Convergence

Theorem. If x is differentiable, then $\lim_{N\to\infty} S_N(x)(t) = x(t)$.

Proof (cont'd). Define $y(\tau) = e^{-j\frac{\tau}{2}}F(\tau)$.

$$\begin{aligned} S_N(x)(t) - x(t) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} F(\tau)\sin\left((N+\frac{1}{2})\tau\right)d\tau \\ &= -\mathrm{Im}\,\frac{1}{2\pi}\int_{-\pi}^{\pi} y(\tau)e^{-jN\tau}d\tau \\ &= -\mathrm{Im}\,\hat{y}[N] \end{aligned}$$

Being continuous on $[-\pi,\pi]$, $y(\tau) \in L_2(2\pi)$. Bessel's inequality implies $\hat{y}[N] \to 0$, so

$$S_N(x)(t) - x(t) = -\mathrm{Im}\,\hat{y}[N] \to 0, \quad \text{as } N \to \infty$$

Dirichlet Test

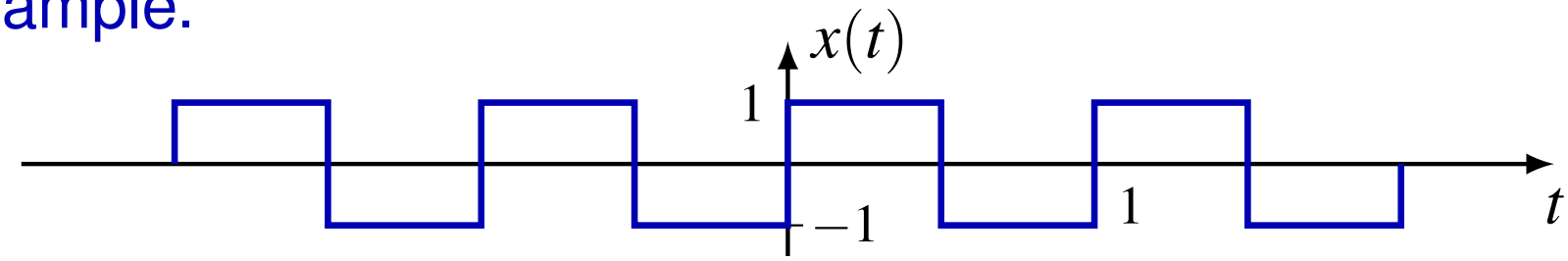
Theorem. If x satisfies following Dirichlet conditions

1. x is bounded, i.e. $\|x\|_\infty < \infty$,
2. x is piecewise monotone, i.e. $\exists$ finite partition of a period s.t. x is monotone on each segment,

then

$$\lim_{N\to\infty} S_N(x)(t) = \frac{x(t_+) + x(t_-)}{2}$$

Example.



$$\lim_{N\to\infty} S_N(x)(t) = \begin{cases} 1, & t \in (n, n+\frac{1}{2}), \\ -1, & t \in (n-\frac{1}{2}, n), \, , \\ 0, & t = \frac{n}{2}, \end{cases} \quad n \in \mathbb{Z}$$

Contents

1. Mean-square Convergence of Fourier Series

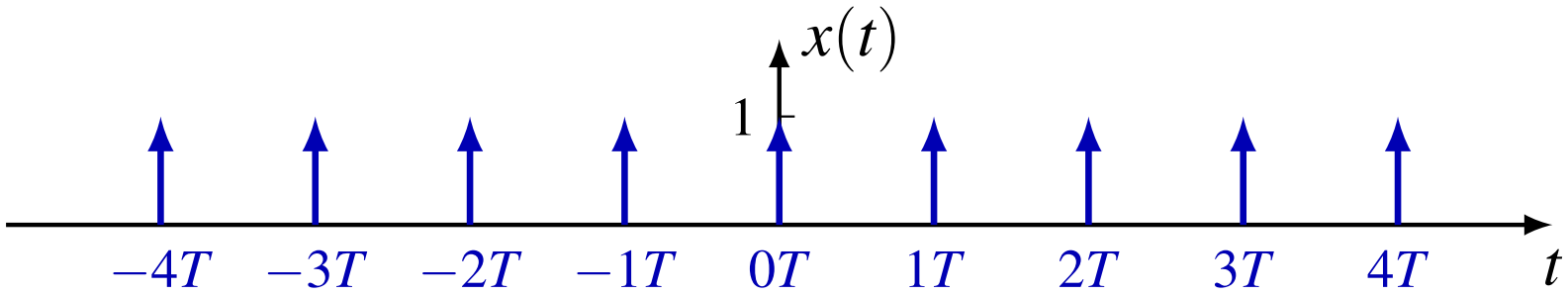
2. Pointwise Convergence of Fourier Series

3. Fourier Series of Periodic Impulse Train

4. Filtering

Periodic Impulse Train

$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$$



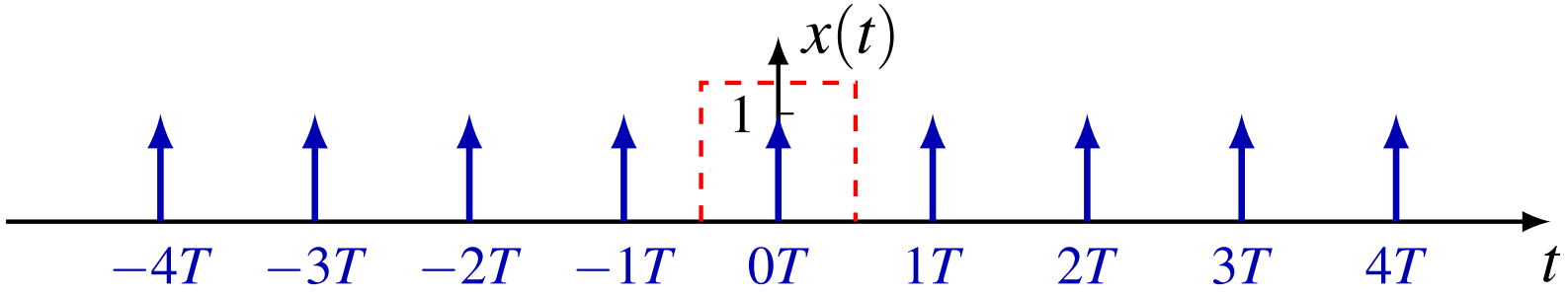
For "good" enough function ϕ, e.g. infinitely differentiable function with compact support,

$$(\phi * x)(t) = \sum_{k=-\infty}^{\infty} \phi(t - kT)$$

$$\int_{\mathbb{R}} \phi(t)x(t)dt = \sum_{k=-\infty}^{\infty} \int_{\mathbb{R}} \phi(t)\delta(t - kT)dt = \sum_{k=-\infty}^{\infty} \phi(kT)$$

Periodic Impulse Train

$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$$



Fourier coefficients

$$\hat{x}[k] = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t) e^{-jk\frac{2\pi}{T}t} dt = \frac{1}{T}$$

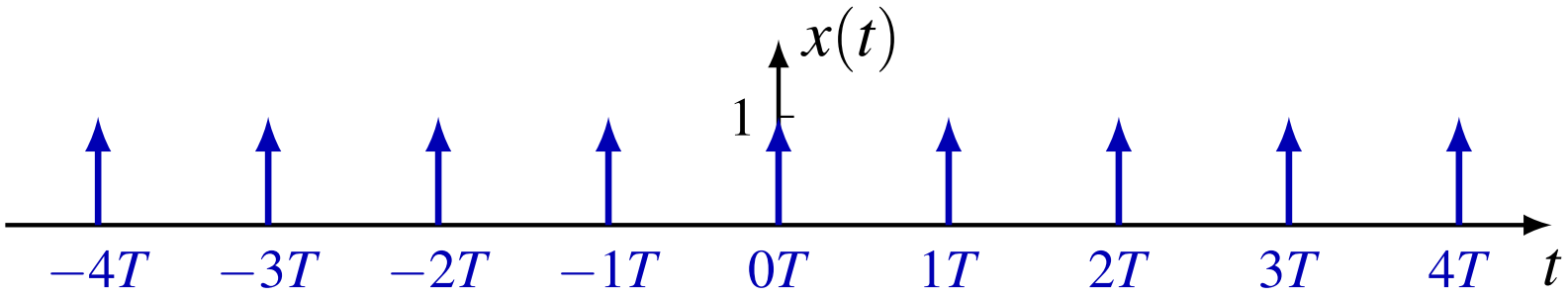
Fourier series

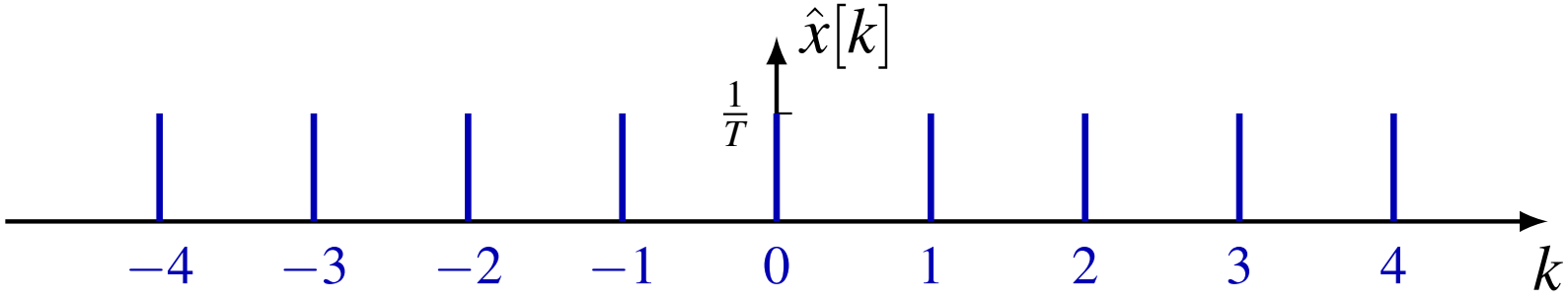
$$x(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T} e^{jk\frac{2\pi}{T}t} = \lim_{N\to\infty} \frac{1}{T} D_N\left(\frac{2\pi}{T}t\right)$$

where D_N is Dirichlet kernel

Periodic Impulse Train

$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT) = \sum_{k=-\infty}^{\infty} \frac{1}{T} e^{jk\frac{2\pi}{T}t}$$





Periodic Impulse Train

$$\int_{\mathbb{R}} \phi(t)x(t)dt = \lim_{N\to\infty} \frac{1}{T}\int_{\mathbb{R}} \phi(t)D_N\left(-\frac{2\pi}{T}t\right)dt \quad (D_N \text{ is even})$$

$$= \lim_{N\to\infty} \frac{1}{2\pi}\int_{\mathbb{R}} \phi\left(-\frac{T}{2\pi}t\right)D_N(t)dt$$

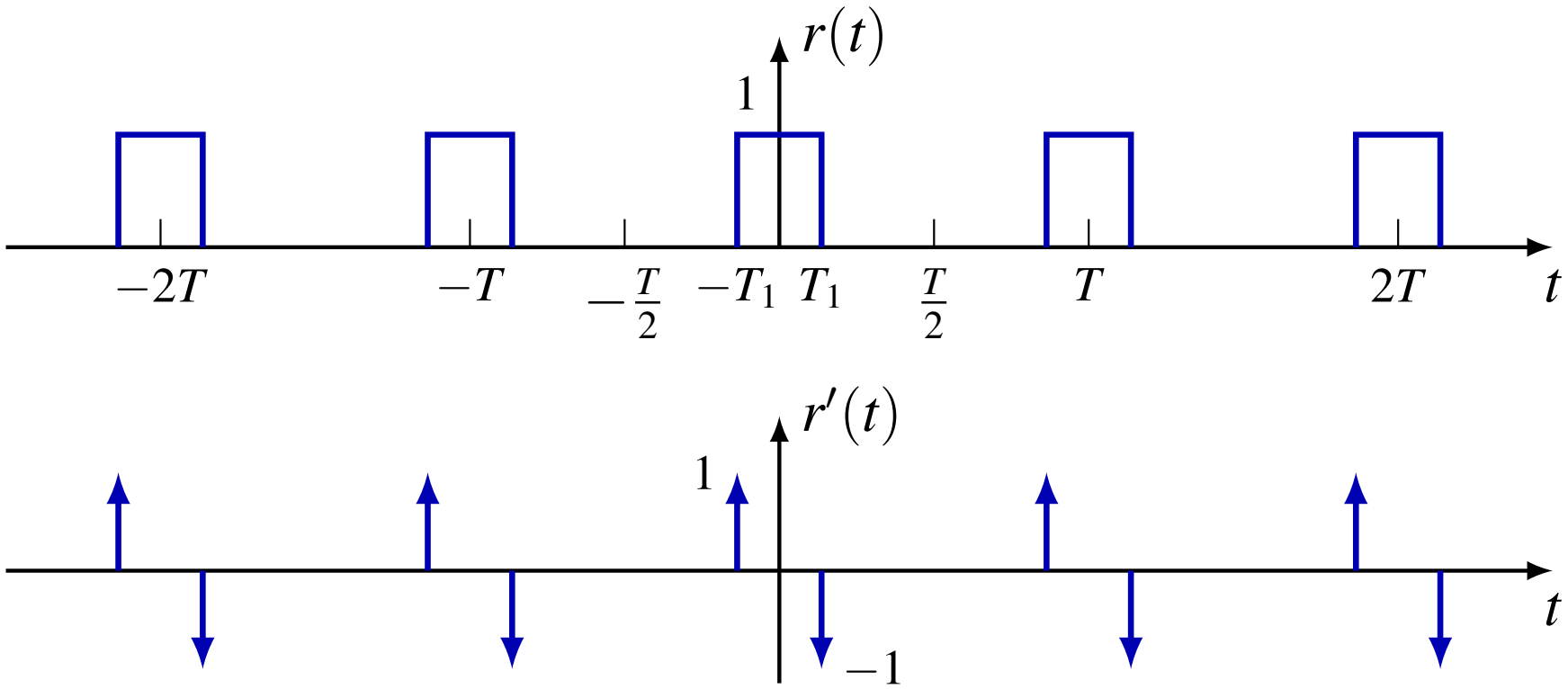
$$= \lim_{N\to\infty} \sum_{k=-\infty}^{\infty} \frac{1}{2\pi}\int_{(2k-1)\pi}^{(2k+1)\pi} \phi\left(-\frac{T}{2\pi}t\right)D_N(t)dt$$

$$= \sum_{k=-\infty}^{\infty} \lim_{N\to\infty} \frac{1}{2\pi}\int_{-\pi}^{\pi} \phi\left(\frac{T}{2\pi}(2\pi k - t)\right)D_N(t-2k\pi)dt$$

$$= \sum_{k=-\infty}^{\infty} \lim_{N\to\infty} \frac{1}{2\pi}\int_{-\pi}^{\pi} \phi\left(\frac{T}{2\pi}(2\pi k - t)\right)D_N(t)dt \quad (D_N \text{ is } 2\pi\text{-periodic})$$

$$= \sum_{k=-\infty}^{\infty} \lim_{N\to\infty} S_N(\phi)(kT) = \sum_{k=-\infty}^{\infty} \phi(kT)$$

Relation with Periodic Square Wave



$$r'(t) = x(t+T_1) - x(t-T_1)$$

$$\widehat{r'}[k] = (e^{jk\omega_0 T_1} - e^{-jk\omega_0 T_1})\hat{x}[k] = \frac{2j\sin(k\omega_0 T_1)}{T}$$

$$\hat{r}[k] = \frac{1}{jk\omega_0}\widehat{r'}[k] = \frac{\sin(k\omega_0 T_1)}{k\pi}, \quad k \neq 0$$

Contents

1. Mean-square Convergence of Fourier Series

2. Pointwise Convergence of Fourier Series

3. Fourier Series of Periodic Impulse Train

4. Filtering

Frequency Response

Recall response of CT LTI system to exponential input e^{st}

$$T\left(\sum_k a_k e^{s_k t}\right) = \sum_k a_k H(s_k) e^{s_k t}$$

where $H(s)$ is system function

$$H(s) = \int_{\mathbb{R}} h(t) e^{-st} dt$$

When restricted to $s = j\omega$, $H(j\omega)$ as function of ω is called frequency response of the system

$$H(j\omega) = \int_{\mathbb{R}} h(t) e^{-j\omega t} dt$$

Filtering

Periodic input x in Fourier series representation

$$x(t) = \sum_{k=-\infty}^{\infty} \hat{x}[k] e^{jk\omega_0 t}$$

Output of LTI system with frequency response $H(j\omega)$

$$y(t) = T\left(\sum_{k=-\infty}^{\infty} \hat{x}[k] e^{jk\omega_0 t}\right) = \sum_{k=-\infty}^{\infty} H(jk\omega_0)\hat{x}[k] e^{jk\omega_0 t}$$

periodic with same periodic, Fourier coefficients related by

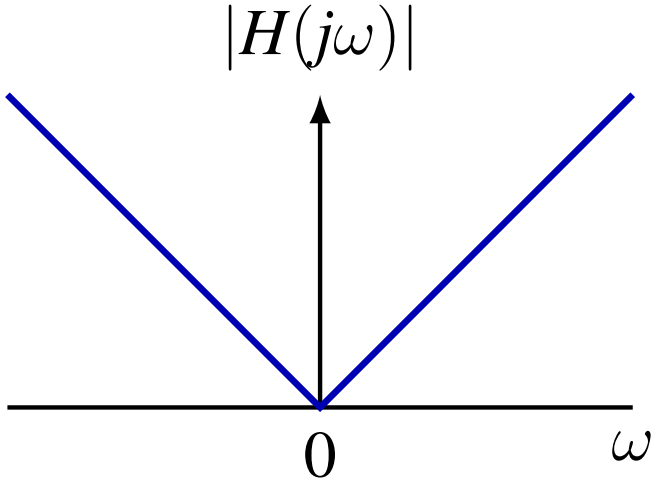
$$\hat{y}[k] = H(jk\omega_0)\hat{x}[k]$$

Filtering changes relative amplitudes of frequency components or eliminates some frequency components entirely

Filtering

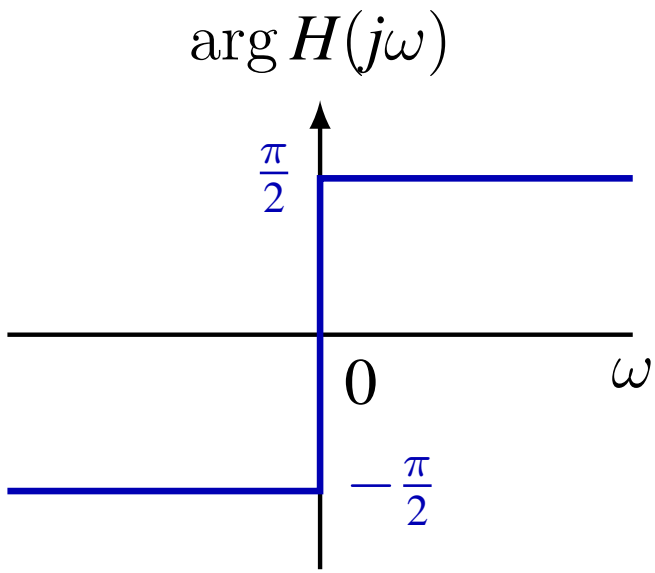
Example. Differentiator $T(x) = x'$ has frequency response

$$H(j\omega) = j\omega$$



For periodic input

$$x(t) = \sum_{k=-\infty}^{\infty} \hat{x}[k] e^{jk\omega_0 t}$$



output

$$y(t) = \sum_{k=-\infty}^{\infty} jk\omega_0 \hat{x}[k] e^{jk\omega_0 t}$$

Fourier coefficients related by $\hat{y}[k] = jk\omega_0 \hat{x}[k]$

Filtering

LTI systems as filters

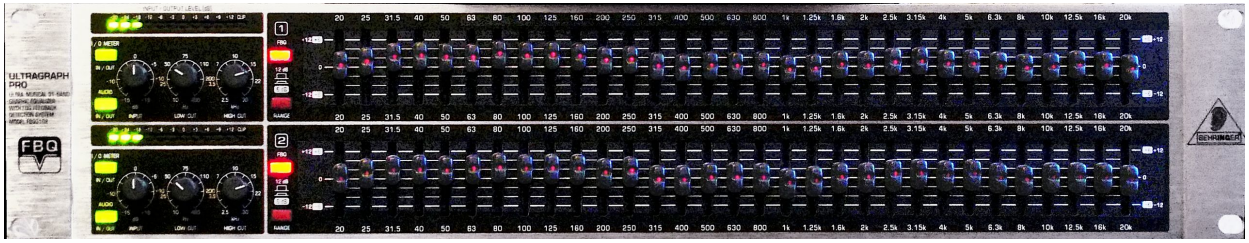
- cannot create new frequency components
- can only scale magnitudes or shift phases of existing components

Example of nonlinear filter: Clipping $y(t) = \max\{x(t), c\}$

Focus on LTI filters in this course

Frequency-shaping filters change shape of spectrum

- e.g. equalizer



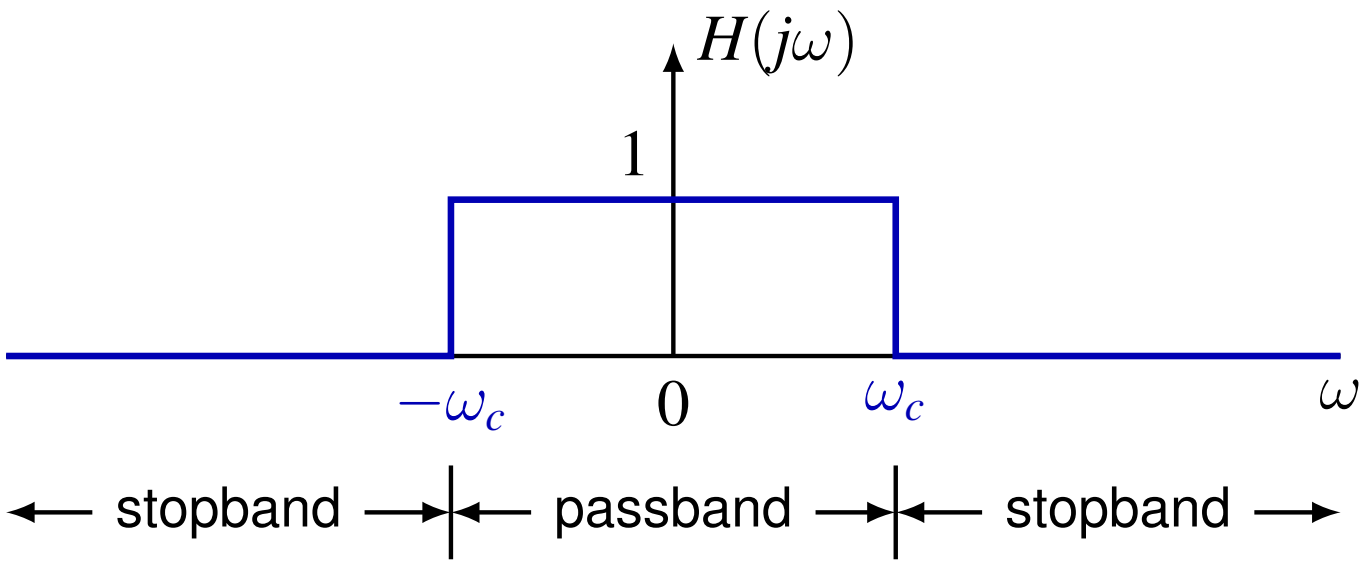
Frequency-selective filters pass some frequencies essentially undistorted and significantly attenuate or eliminate others

Ideal Frequency-selective Filters

Ideal lowpass filter

$$H(j\omega) = \begin{cases} 1, & |\omega| \leq \omega_c \\ 0, & \text{otherwise} \end{cases}$$

ω_c: cutoff frequency

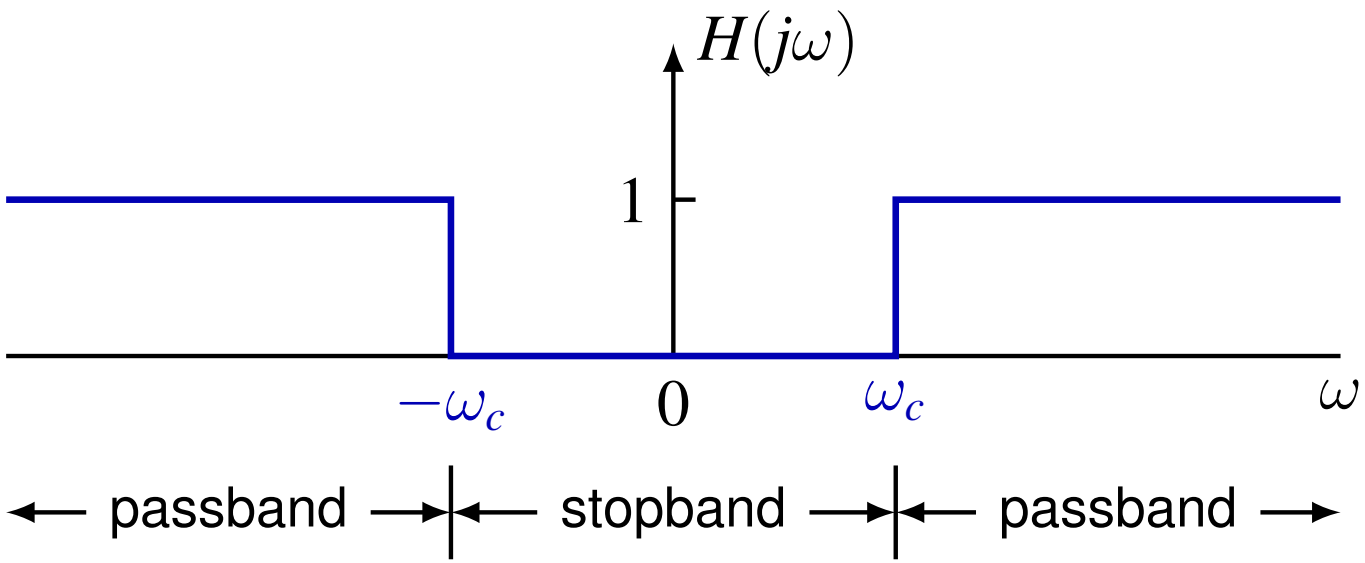


Ideal Frequency-selective Filters

Ideal highpass filter

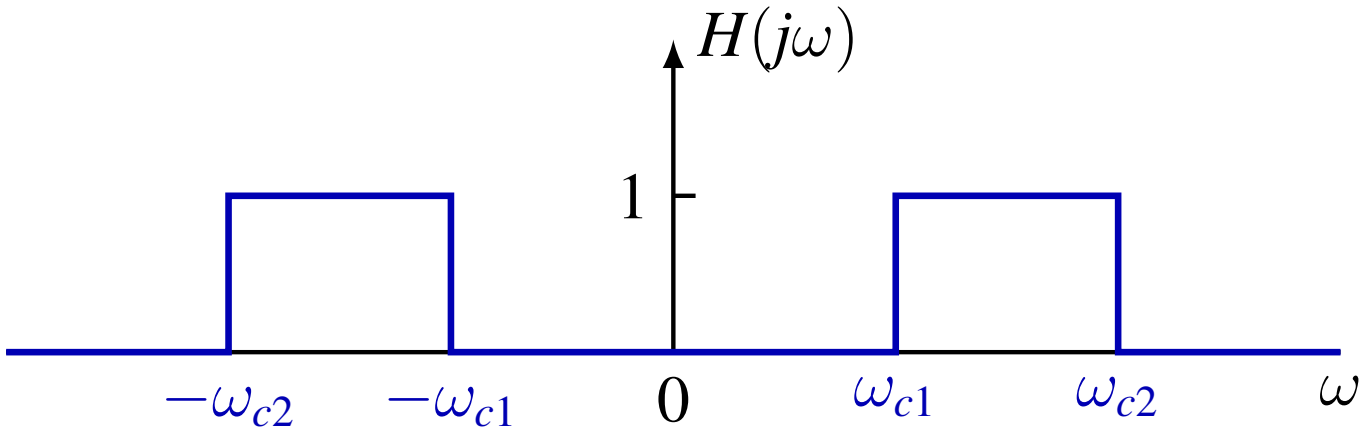
$$H(j\omega) = \begin{cases} 1, & |\omega| \geq \omega_c \\ 0, & \text{otherwise} \end{cases}$$

ω_c: cutoff frequency



Ideal Frequency-selective Filters

Ideal bandpass filter

$$H(j\omega) = \begin{cases} 1, & \omega_{c1} \leq |\omega| \leq \omega_{c2} \\ 0, & \text{otherwise} \end{cases}$$

ω_{c1}: lower cutoff frequency
ω_{c2}: upper cutoff frequency